

UNIT: 3 OUTPUT DEVICES

1. WHAT IS OUTPUT DEVICE.

- The output devices are the devices which are used to display the result generated by the computer system.
- Monitor, printer, plotter, speaker are the example of output devices.

FUNCTIONS OF INPUT DEVICES:

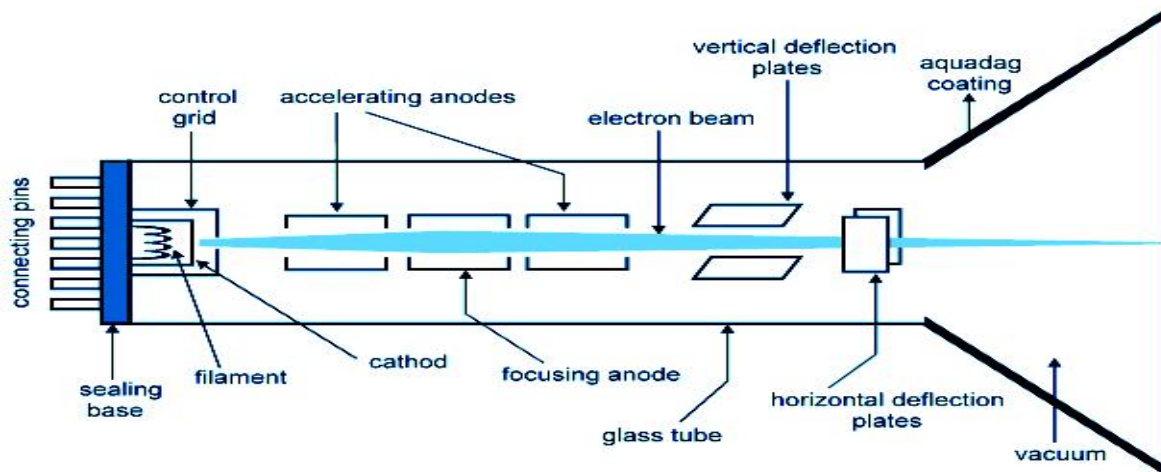
- Accept the result from the CPU.
- Convert that result into human readable form.
- Supply this result to output device.

2. EXPLAIN THE VISUAL DISPLAY UNIT

OR

EXPLAIN CRT (CATHOD RAY TUBE) MONITER.

- The monitor is the common output device mostly used It is a softcopy output device.
- It can be thought of as a high resolution TV set.
- The monitor can also determine if the display will be colour, black and white, or include graphical objects (pictures).
- Two types of monitors are used.
 - CRT monitors.
 - Non CRT monitors.
- Most computer monitors are based on Cathode Ray Tube (CRT) technology.
- The basic operation of these tubes is similar to that in television sets.



- In CRT display CRT is a specialised vacuum tube in which images are produced when electron beam strikes a phosphor surface.
- CRT monitor contains cathode, control grid, acceleration anode, deflection plates & phosphor coated screen.
- Cathode: the cathode is heated by filament and produced high speed & large amount of electrons.
- Control Grid: used to control the brightness of the screen. It controls the number of electrons.
- Accelerating anodes: they are with focusing lens are applied with positive electrons.
- Horizontal deflection plate: moves electron side by side.
- Vertical deflection: moves electrons up & down.
- Screen: contains millions of tiny red, green, blue phosphor dot that glow when struck by electron beam that travels across screen to create a visible image.

ADVANTAGES OF CRT DISPLAY

- Produce more colours.
- Price is lower than LCD & Plasma.
- High contrast ratio.
- Can easily increase brightness of monitor by reflecting the light.

DISADVANTAGES OF CRT DISPLAY

- High power consumed.
- Heavy to pick up and carry.
- Large space required.

3. EXPLAIN THE NON CRT DISPLAY.

LCD (Liquid Crystal Display)

- In LCD, a liquid crystalline material is sandwiched between two glass or a plastic plates.

- The front plate is transparent and the back plate is reflective.
- There is a coating of thin film on the front plate.
- The coating is transparent and conductive. Its sections (segments) are in the shape of desired characters.
- LCDs do not emit their own light. Therefore, a light source is to be used.
- LCDs simply change the reflection of available light. Today, most LCDs used are of the type that produces dark images on a silver background.

Advantages:

- Light weight as compare to CRT.
- Perfectly flat screen.
- Consumed low electricity power.
- Able to generate higher brightness in images.

Disadvantages:

- Fixed resolution that cannot be changed.
- Expensive than CRT display.
- Limited viewing angle.
- Short life.

PDP (Plasma Displays Panel)

- In Plasma Displays, ionized gas is sandwiched between two glass plates.
- A number of parallel wires run horizontally as well as vertically.
- A small amount of current is passed through one horizontal and one vertical wire to cause the gas to glow at a spot at the intersection of the wires.
- The IBM 581 display employs 960 horizontal and 768 vertical pixel as compared to IBM-PC colour graphic adapter which is provided with 320 X 200 pixels in medium resolution and 640 X 200 in high resolution.

Advantages:

- Large viewing angle.
- Thinner in width.
- Free standing or can be easily mounted on wall.
- Clear image, brighter viewing angle, better colour quality & high contrast ratio.

Disadvantages:

- The plasma displays screens are costly.
- These are available on the selected models of portable computers.
- More electricity than LCD.

- As your plasma get older the brightness get dimmer.

4. WHAT IS PRINTER? EXPLAIN THE TYPES OF PRINTER.

Printer

- The printer is a most commonly used output device.
- It is used to producing the hard copy output.
- It prints characters, symbols & graphics on the paper.
- Printer can be categorised according to the technology used in printer, speed, and approach of printing, colours, language & the quality of printing.
- Mainly printer can be classified in two types:
 - Impact printer
 - Non impact printer

IMPACT PRINTER:

- It works on the same mechanism of type-writer.
- It forms a character or image by striking mechanism such as hammer or wheel against to ink ribbon, leaving an image on paper.
- It is oldest technology and still is in used.
- It can capable to print single character or line at the same time.
- Commonly types of impact printers are dot matrix, daisy wheel, chain, drum printer.

CHARACTERISTICS OF IMPACT PRINTER:

- Physical contact with paper to produce output.
- Low cost
- Very noisy
- Very slow in printing
- Low quality print out
- Stand with dusty or extreme environment

NON IMPACT PRINTER:

- Non impact printer forms characters & images on paper without actually striking the paper.
- Paper & print head come in contact & hence the text or image is formed.
- Ink jet & laser printer are example of non impact printer.

CHARACTERISTICS OF NON IMPACT PRINTER

- Faster than impact printer.

- Ability to change type face automatically.
- High quality output.
- Support transparency.
- More expensive than impact printer.
- Less maintenance than impact printer.

5. EXPLAIN IMPACT PRINTERS.

Dot matrix printer:

- Character printer.
- Capable to print single character at the same time.
- Forms characters & images as a pattern of dots.
- Contains a print head which moves horizontally across paper.
- Uses 5 × 7 matrix to form a character.
- Print by hammering the pins on inked ribbon to leave ink impressions on the paper.
- **Able to print 30 to 600 characters per second.**
- **ADVANTAGES:**
 - Low cost & easily available.
 - Cheap in cost.
 - Can make carbon copy of print out.
 - Low maintenance cost.
 - Work with any type of environment.
- **DISADVANTAGES:**
 - Slow in speed.
 - Very noisy.
 - Cannot work perfectly in graphics.

DAISY WHEEL PRINTER:

- Character printer.
- Able to print a single character at the same time.
- Contain a metal wheel on which the characters & numbers are raised on the each petal.
- The wheel is rotated very fast when the desired characters arrives at correct position a print hammer strike to produce output.
- Different type of font face can be used by replacing the daisy wheel.
- Able to print bold letter by striking on specific characters twice or thrice.

- **Capable to print 10 to 50 characters per second.**
- **ADVANTAGES:**
 - Low cost.
 - Can make carbon copy of print out.
 - Low maintenance cost.
 - Printing quality is similar to a type writer.
 - Able to print bold characters.
 - Allows using different font-face in same document.
- **DISADVANTAGES:**
 - Very slow in speed.
 - Very noisy.
 - Cannot print graphics.

DRUM PRINTER:

- It's a line printer.
- Able to print a line at the same time.
- Consist of a solid cylindrical drum with characters embossed on it in circular band.
- Each band consists of character set which contains 96 characters.
- Drum rotates fastly when desired characters arrives an appropriate hammer stike on ribbon & character is print on paper.
- **Capable to print 300 to 2000 lines per minute.**
- **ADVANTAGES:**
 - Low cost than non impact printer.
 - Can make carbon copy of print out.
 - Low maintenance cost.
 - Faster than other impact printer.
 - Printing quality is similar to a type writer.
- **DISADVANTAGES:**
 - Very slow in speed.
 - Very noisy.
 - Large & heavy.
 - Cannot print graphics.
 - Only prints predefined set of characters.

CHANIN PRINTER:

- It's a line printer.

- Able to print a line at the same time.
- Consist of a metallic chain on which all characters of character set are embossed.
- Character set contains 48, 64 or 96 characters.
- Characters are embossed several times.
- Chain rotates at high speed when the desired characters in correct position the hammer strikes & the characters are print on paper.
- Capable to print 400 to 2500 lines per minutes.
- ADVANTAGES:
 - Low cost than non impact printer.
 - Can make carbon copy of print out.
 - Chain can be easily changed.
 - Allowed to print different type font.
 - Printing quality is similar to a type writer.
- DISADVANTAGES:
 - Slower than non impact printer.
 - Very noisy.
 - Large & heavy.
 - Cannot print graphics.
 - Only prints predefined set of characters.

7. EXPLAIN NON-IMPACT PRINTERS.

INK-JET PRINTER:

- It's non-impact printer.
- It's a character printer.
- Forms characters and all kinds of images by spraying drops of ink on to the paper.
- Print head contains 64 tiny nozzles.
- To print a character the printer the printer selectively heats the appropriate set of nozzle as the print head moves horizontally.
- Inkjet printer can either colour or monochrome.
- Capable to print 30 to 400 characters per minutes.
- ADVANTAGES:
 - High quality output.
 - Silent during the operation.
 - Able to print graphics.
 - Able to print any characters & graphics.

- Able to generate colour & monochrome output.
- **DISADVANTAGES:**
 - Slower than dot matrix printer.
 - Cannot make carbon copy of print out.
 - Expensive than impact printer.

LASER PRINTER:

- It's non-impact printer.
- It's a page printer.
- Three main components laser beam, a multi-sided mirror, a photoconductive drum & toner.
- To print page laser beam is focused on drum by spinning multisided mirror.
- Drum is electric charged.
- Toner which is composed of oppositely charged ink particles, stick to the drum.
- Then toner focused on the paper with heat & pressure to generate output.
- **Low speed laser printer can print 4 to 12 page per minute while high speed laser printer Capable to print 500 to 1000 pages per minutes.**
- **ADVANTAGES:**
 - High quality output.
 - Very faster in speed.
 - Silent during the operation.
 - Able to print graphics.
 - Able to print any characters & graphics.
 - Able to generate colour & monochrome output.
- **DISADVANTAGES:**
 - Very expensive.
 - Cannot make carbon copy of print out.

8. WRITE DOWN THE DIFFERENCE BETWEEN IMPACT & NON-IMPACT PRINTER.

Impact printer	Non impact printer
Printing character by striking hammer against ink ribbon to produce output	Printing characters or graphics by spraying ink on paper.
Slow in speed	Faster than impact printer
Work with any environment	Can't work with all environment
Less expensive than non impact printer	More expensive than impact printer

Noisy during printing	Silent during printing
Able to produced carbon copy output	Cant able to produced carbon copy output
e.g.: daisy wheel, drum, chain, dot matrix	e.g.: inkjet, laser

9. WHAT IS PLOTTER? EXPLAIN ITS TYPE.

Plotter:

- Plotter is an output device which is capable to producing hardcopy output of graphics.
- Used to producing wide format printing.
- It is an ideal output device for architects, engineers, city planners and other who need to generate hardcopy output of widely varying in sizes.
- Woks on the same mechanism of human holding pen & moving on paper.
- Contains multiple pens & pencil which can be easily changed out in order to create drawing of different colours.
- Normally generate output very slowly.
- Mainly two types of plotters are used:
 - Drum plotter
 - Flatbed

Drum plotter:

- In the case design has to be made is placed over a drum.
- It consist one or more than penholders which are mounted to the drum surface.
- The drum plotter both the paper and the pen move.
- The paper is contained on two rollers and passes over a drum.
- The pen is driven along fixed arm set across the length of the drum.
- It is especially useful for plotting continuous line graph.
- The accuracy of the drum type depends on the paper transport mechanism, which in turn is dependent on the width of the paper and it can also produce larger drawings.

Flatbed Plotter

- The Flatbed Plotter is generally more expensive and can produce very detailed and accurate drawings.
- The paper is mounted on a stationary flatbed.

- The pen is mounted on a moveable arm.
- Colour drawing can be produced by some plotters through interchangeable pens.
- Able to draw output in small size as A4 size or can able to generate very large size can be up to 20ft by 50ft.
- Specially used in the design of cars, ships, aircrafts, buildings, highways etc.

9. WRITE A BRIEF NOTE ON SPEAKER.

- The speaker is output device which is connected to computer's soundcard.
- The speaker output the sound generated by the sound card.
- Audio data is generated by computer is send to audio card which is located inside extension slot.
- It can translate data into audio signal which are sending to speaker.
- The speaker can able to produced series of different tones.